

Aim

To train a Neural Network using TensorFlow and evaluate Logistic Regression using TensorFlow.

Objective

To study:

- 1. TensorFlow framework**
 - 2. Neural Network training process**
 - 3. Logistic Regression model**
 - 4. Data preprocessing and model evaluation**
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Theory

What is TensorFlow?

TensorFlow is an open-source machine learning and deep learning framework developed by Google.

It is mainly used for:

- Deep learning**
- Artificial Intelligence**
- Neural Networks**
- Data analysis**
- Image processing**
- Natural Language Processing**

TensorFlow provides powerful libraries for building and training neural network models efficiently.

It supports multiple programming languages such as:

- Python**
- Java**
- JavaScript**
- C++**

TensorFlow 2.0 was released in 2019 with improved features and easy implementation using Keras API.

What is PyTorch?

PyTorch is another popular deep learning framework mainly used for:

- Research**
- Neural networks**

- AI applications
- Deep learning models

It is based on Python and Torch library.

PyTorch is widely used because:

- It is easy to understand
 - Dynamic computation graph is supported
 - Faster debugging is possible
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What is Regression?

Regression is a machine learning technique used to predict continuous values.

Examples:

- Salary prediction
- Age prediction
- Stock price prediction

Regression finds relationship between:

- Independent variables (inputs)
 - Dependent variables (outputs)
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What is Logistic Regression?

Logistic Regression is a supervised machine learning algorithm used for binary classification problems.

It predicts probability values between:

- 0
- 1

Examples:

- Cancer detection
- Spam email detection
- Pass/Fail prediction

The logistic regression output is calculated using sigmoid function.

Sigmoid Function:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Where:

- Output near 1 → Positive class

- Output near 0 → Negative class
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What is Neural Network?

A Neural Network is a computational model inspired by the human brain.

It consists of:

1. Input Layer
2. Hidden Layer
3. Output Layer

Each neuron processes information and passes output to next neuron.

Neural networks are used in:

- Image recognition
 - Speech recognition
 - Robotics
 - Medical diagnosis
 - NLP
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Dataset Used

The program uses Breast Cancer Dataset from Scikit-learn library.

The dataset contains:

- Tumor features
 - Cancer labels:
 - Malignant
 - Benign
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Working of Program

1. Load dataset
 2. Split dataset into training and testing data
 3. Normalize input data using StandardScaler
 4. Build neural network model
 5. Train logistic regression model
 6. Evaluate model accuracy
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Code

```
import tensorflow as tf

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.datasets import load_breast_cancer

# Load the breast cancer dataset
data = load_breast_cancer()

X_train, X_test, y_train, y_test = train_test_split(
    data.data,
    data.target,
    test_size=0.2,
    random_state=42
)

# Scale the input features
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Build the model
model = tf.keras.models.Sequential([
    tf.keras.layers.Dense(
        1,
        activation='sigmoid',
        input_shape=(X_train.shape[1],)
    )
])

# Compile the model
model.compile(
```

```
optimizer='adam',  
loss='binary_crossentropy',  
metrics=['accuracy']  
)  
  
# Train the model  
model.fit(X_train, y_train, epochs=5)  
  
# Evaluate the model  
test_loss, test_accuracy = model.evaluate(X_test, y_test)  
  
print("Accuracy:", test_accuracy)
```